

A Q-THEORETIC SOLUTION TO
A'INGAE POSTLABIAL ROUNDING
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Abstract: I describe and analyze a process of postlabial rounding in A'ingae (isolate, ISO 639-3: con): after labial consonants, the diphthong /ae/ may surface as [oe]. However, a postlabial monophthongal /a/ always surfaces faithfully as [a]. To capture these facts, I propose an analysis couched in Q-Theory, where one vocalic target of a diphthong corresponds to fewer subsegments than a monophthong. This predicts that diphthongs might show an emergence-of-the-unmarked (TETU) effect, while monophthongs surface faithfully. The prediction is borne out by A'ingae postlabial rounding, contributing a novel argument for the Q-Theoretic representations. Finally, I show that similar asymmetries between diphthongs and monophthongs have recurred in the language's history, further strengthening the proposal.

Keywords: diphthong, labialization, TETU, Cofán, assimilation, subsegment

1 Introduction

This squib describes and analyzes the A'ingae process of postlabial rounding. In A'ingae (or Cofán, an Amazonian isolate, ISO 639-3: con), the underlying sequence /ae/ optionally rounds to [oe] after labial consonants ($p\ p^h\ mb\ fv\ m$). While postlabial rounding has been reported in previous literature (e.g., Lakshmi 1982, Galloway 1990, Lionnet 2017), the A'ingae pattern is unusual in that it is restricted to a diphthong; the monophthongal /a/ never rounds to $^\times[o]$ in the same postlabial environments.

To account for the fact that the postlabial rounding may apply to /ae/, but not /a/, I adopt a Q-Theoretic (e.g., Inkelas and Shih 2016, 2017, Garvin, Lapierre, and Inkelas 2018) representation of vowels.

I would like to wholeheartedly thank members of the Cofán communities whose generosity and assistance have made this research uniquely possible. Thanks in particular to Jorge Criollo and his family for welcoming me to their home; my primary collaborators on this project, Jorge Mendúa, Marcelo Lucitante, and Shen Aguinda, for their insight, patience, and kindness; as well as Hugo Lucitante for support with all matters along the way. I would also like to thank Hannah Sande, Karee Garvin, Chelsea Sanker, Scott AnderBois, and the audiences at 29mfm (29th Manchester Phonology Meeting), Phorum (Berkeley Phonetics, Phonology, and Psycholinguistics Forum), and ALDP (Brown University A'ingae Language Documentation Project) for helpful discussions and their invaluable feedback.

This project was supported in part by National Science Foundation 20-538 Linguistics Program's Doctoral Dissertation Research Improvement grant 2314344 for *Doctoral Dissertation Research: Nominal and deverbal morphology in an endangered language*, American Philosophical Society's Lewis and Clark Fund for Exploration and Field Research on *Stress and glottalization across lexical classes in A'ingae (or Cofán; Ecuador)*, and California Language Archive's Oswald Endangered Language Grants for fieldwork research on the following projects: *Phonology-syntax interface in A'ingae*, *Documenting the A'ingae noun phrase*, and *Form and productivity in A'ingae derivation*.

I assume that the monophthongal *a* consists of three subsegments ($a^1 a^2 a^3$), but the *a*-component of a diphthong consists of only two ($a^1 a^2 e^2 e^3$). As a result, one vocalic target of a diphthong is weaker (i.e., corresponds to fewer subsegments) than a monophthong (Schwarz et al. 2019, Garvin et al. 2020). Since each subsegment may be independently subject to feature IDENTITY constraints, Q-Theory predicts that diphthongs might show emergence-of-the-unmarked (TETU) effects (McCarthy and Prince 1994), while monophthongs surface faithfully. This prediction is borne out by A'ingae postlabial rounding, contributing a novel argument for the subsegmental representations of Q-Theory. (The process does not apply to other diphthongs, including /ai/. A specific account of the underapplication of postlabial rounding is beyond the scope of this squib.)

The rest of the squib is structured as follows. Section 2 gives background on the language and its speakers. Section 3 presents basic aspects of the A'ingae phonemic inventory and describes the postlabial rounding of /ae/. Section 4 proposes a Q-Theoretic analysis of the process. Section 5 discusses the postlabial diachrony of other diphthongs. Section 6 concludes.

2 Language Background

A'ingae (or Cofán, ISO 639-3: con) is an Amazonian isolate (Ander-Bois et al. 2019, Hammarström et al. 2020; cf. Rivet 1924, 1952, Ruhlen 1987) with about 1,500 native speakers in the northeast Ecuadorian province of Sucumbíos and the southern Colombian department of Putumayo. The language is endangered and under severe socioeconomic pressures. Despite challenges, language attitudes among the Cofán are uniformly positive (Dąbkowski 2021a).

Previous work on A'ingae includes Repetti-Ludlow et al.'s (2019) and Brandt's (2024) phonetic studies, Borman's (1962) and my (Dąbkowski 2024a) phonological descriptions, my (Dąbkowski 2023) diachronic account of postlabial raising, Sanker and AnderBois's (2024) internal reconstruction of nasality, Repetti-Ludlow's (2021) analysis of laryngeal co-occurrence restrictions, my (Dąbkowski 2021b, 2024b, to appear) work on stress and glottalization, and Fischer and Hengeveld's (2023) grammatical sketch.

The data presented in this squib represent the Ecuadorian language variety and reflect the judgments of three native-language consultants from Dureno and Sinangoé, Sucumbíos. All the fieldwork data have been deposited in the California Language Archive as Dąbkowski 2020. Section 5 incorporates data from Borman's (1976) dictionary, which reflects the language as spoken 50–70 years ago.

3 Description

The A'ingae syllable structure can be schematized as (C)V(V)(?), with optional consonantal onsets, maximally diphthongal nuclei, and optional coda glottals (Dąbkowski 2024a). The language's phonemic inventory is moderately large, totaling 27 consonants (table 1) and 5

Table 1
A'ingae consonants

	Labial	Alveolar	Sibilant	Postalveolar	Velar	Glottal
Plain stop	<i>p</i>	<i>t</i>	<i>ts</i>	<i>tʃ</i>	<i>k</i>	<i>ʔ</i>
Aspirated stop	<i>p^h</i>	<i>t^h</i>	<i>ts^h</i>	<i>tʃ^h</i>	<i>k^h</i>	
Prenasal stop	<i>^mb</i>	<i>ⁿd</i>	<i>ⁿdʒ</i>	<i>ⁿdʒ̥</i>	<i>^ŋg</i>	
Fricative	<i>f</i>		<i>s</i>	<i>ʃ</i>		<i>h</i>
Oral sonorant	<i>v</i>	<i>r</i>		<i>j</i>	<i>ɰ</i>	
Nasal stop	<i>m</i>	<i>n</i>		<i>ɲ</i>		

Table 2
A'ingae monophthongs

	Front	Central	Back
High	<i>i, ĩ</i>	<i>i, ĩ</i>	
Low	<i>e, ē</i>	<i>a, ā</i>	<i>o, ō</i>

Table 3
A'ingae diphthongs

<i>ie, iē</i>	<i>ii, iĩ</i>	<i>io, iō</i>
<i>ei, ēi</i>	<i>oe, ōē</i>	<i>oi, ōi</i>
<i>ia, iā</i>		<i>oa, ōā</i>
<i>ai, āi</i>	<i>ae, āē</i>	<i>ao, āō</i>

monophthongal vowels (table 2) (Borman 1962, Repetti-Ludlow et al. 2019). Phonetically, the A'ingae vowels have the following properties: *i* is high and front; *ĩ* is high and central (to back); *e* is (high-)mid and front; *a* is central and low; and *o* is back, high-mid (to high), and rounded (Repetti-Ludlow et al. 2019, Brandt 2024). The 5 vowel qualities form 11 contrastive diphthongs (table 3) (Dąbkowski 2023, 2024a). Both monophthongs and diphthongs contrast oral and nasal articulations; diphthongs are either fully oral or nasal.

Word-initially (1a) and after a nonlabial consonant (1b–e), all of the A'ingae diphthongs are realized faithfully, including /ae/ (1d–e).¹

- (1) *Faithful realization of diphthongs in most contexts*
- | | | | | |
|---------|-------------------------|------------------------|--------------|-----------|
| a. /āi/ | b. /tʃ ^h oi/ | c. /k ^h ii/ | d. /ānāēʔmā/ | e. /ākāē/ |
| [āi] | [tʃ ^h oi] | [k ^h ii] | [ānāēʔmā] | [ākāē] |
| dog | row | lie.down | hammock | install |

However, after a labial consonant the diphthong /ae/ is optionally (though often) rounded to [oe] (Dąbkowski 2024a). The process is exceptionless in that all instances of postlabial /ae/ can (but need not) be rounded. Rounded realizations ([oe]) predominate in regular and

¹ The following glossing abbreviations have been used: ADV = adverbial, CAUS = causative, PRD = periodic.

fast speech. Unrounded realizations ($[ae]$) can be heard in careful or drawn-out pronunciation. In other words, the postlabial rounding of $/ae/$ is speech-rate-dependent. This variability finds a reflection in written language, with speakers choosing alternately “*ae*” or “*oe/ue*” (depending on the orthography) to represent the postlabial diphthong. Postlabial rounding is observed in roots (2) as well as derived environments (3) and is unaffected by nasality, with both $/ae/$ and $/āē/$ optionally rounded to $[oe]$ ((2a–c), (3d)) and $[ōē]$ ((2d), (3a–c)). An unrounded realization of (2a) is given in figure 1. A rounded realization is given in figure 2.

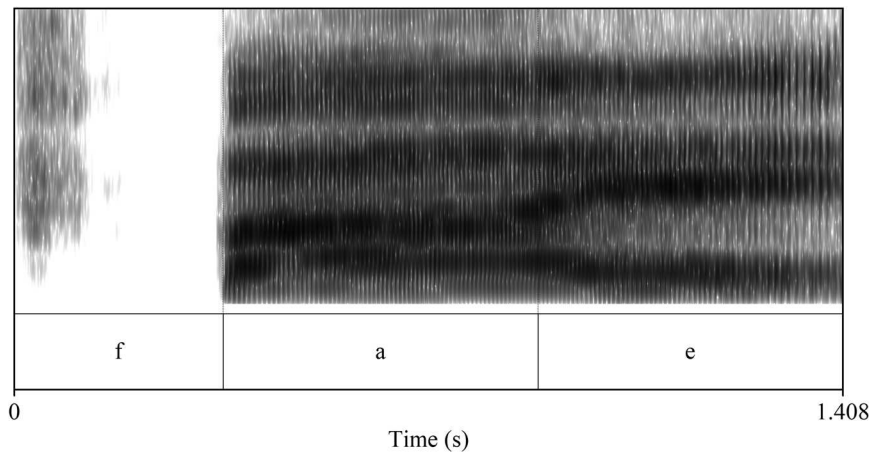


Figure 1
 Unrounded realization of $/fae/$ ‘one’

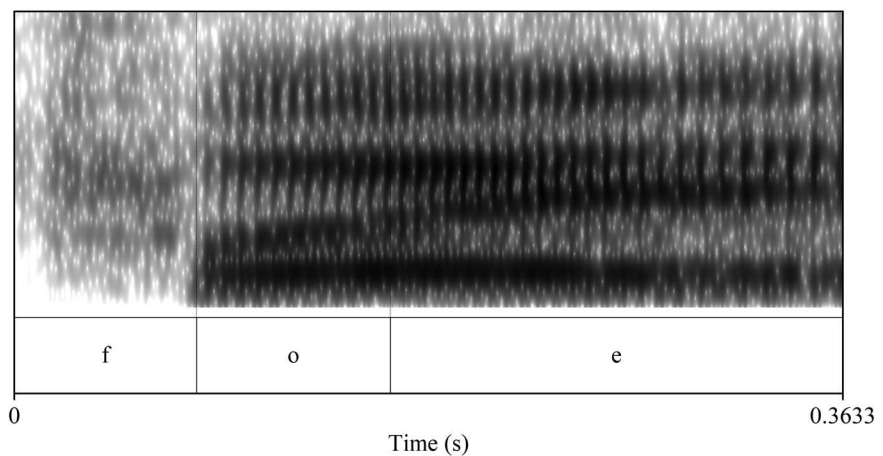


Figure 2
 Rounded realization of $/fae/$ ‘one’

(2) *Rounding of /æ/ after labials morpheme-internally*

a. /fae/	b. /faesi/	c. /vaeji/	d. /māē/
[fae]	[faesi]	[vaeji]	[māē]
[foe]	[foesi]	[voeji]	[mōē]
one	other	recently	command

(3) *Rounding of /æ/ after labials across morpheme boundaries*

a. /atapa-ē/	b. /nā ^m ba-ē/	c. /sī [?] mā-e/	d. /tse [?] fa-e/
[atapāē]	[nā ^m bāē]	[sī [?] māē]	[tse [?] fae]
[atapōē]	[nā ^m bōē]	[sī [?] mōē]	[tse [?] fae]
breed-CAUS	turbid-CAUS	bruised-ADV	that.side-ADV

The monophthongal /a/ does not undergo rounding after labial consonants and is always faithfully realized as [a] (4).

(4) *No rounding of monophthongal /a/ after labials*

a. /pa/	b. /fā [?] fā/	c. /va/	d. /mā [?] p ^h ā/
[pa]	[fā [?] fā]	[va]	[mā [?] p ^h ā]
die	diarrhea	this	wash

4 Analysis

Postlabial rounding is an instance of assimilation, whereby the labial feature of a consonant “spreads” onto the following vowel. I preliminarily capture the process with the LABIALASSIMILATION constraint (5). (LABIALASSIMILATION will later be revised to refer to Q-Theoretic sub-segments.)

(5) *LABIALASSIMILATION (PAssM)* (to be revised)

Assign a violation mark for each consonant immediately followed by a vowel with a different [±Labial] value.

Gratuitous labialization is prevented by the opposing force of input-output faithfulness. Specifically, I use a relativized IDENTITY constraint, which penalizes output candidates that differ from the input with respect to a specific feature. Following Brandt (2024), I assume featural specifications for the A’ingae vowels as given in table 4. Since *a* and *o* differ only in the feature [±Labial], the constraint relevant to postlabial rounding is IDENTITY[LABIAL] (6).

(6) *IDENTITY[LABIAL] (IDENTP)*

Assign a violation mark each time the feature [±Labial] has a different value in the input than in the output.

Table 4
A’ingae vowel features

	<i>i</i>	<i>ĩ</i>	<i>e</i>	<i>a</i>	<i>o</i>
High	+	+	—	—	—
Front	+	—	+	—	—
Labial	—	—	—	—	+

Source: Based on Brandt 2024

To capture the optionality in A'ingae postlabial rounding, I adopt a weighted-constraint model of the grammar (Smolensky and Legendre 2006). The relative counts of rounded and unrounded postlabial realizations of /ae/ are unavailable. In the analysis to be presented, I assume that they are equally frequent (although using any other ratio would not affect the core claim of the proposal).

Now, the markedness constraint PASSM penalizes sequences of a labial consonant plus an unrounded vowel, but its activity is revealed only in its effect on the diphthong /ae/; monophthongs are unaffected. This is a challenge because—unless something more is said about the difference between monophthongs and diphthongs—the constraint PASSM targets the two equally. This is to say, if PASSM has a weight high enough to correctly predict that diphthongs undergo optional rounding, it will incorrectly predict that monophthongs should also be able to round after labials (table 5). If, on the other hand, PASSM has a weight low enough to predict monophthongal faithfulness, it fails to predict the possibility of diphthongal rounding (table 6).

To capture both the postlabial rounding seen in diphthongs and its underapplication to monophthongs, I adopt the subsegmental representations of Q-Theory (Inkelas and Shih 2016). Q-Theory posits that each segment (Q) consists of multiple—most commonly three—subsegments (q) representing closure (q¹), hold (q²), and release (q³). In segments with one articulatory target, the three q's are identical. For example, the low vowel segment (Q) *a* is represented with three subsegments as (a¹ a² a³). Internally complex segments have multiple articulatory targets. In Q-Theory, the different targets are mapped on to different q's. The affricate *ts*, for example, may be represented as (t¹ s² s³) (Inkelas and Shih 2017). The circumoralized nasal ^b*m*^b may be represented as (b¹ m² b³) (Garvin, Lapierre, and Inkelas 2018, Garvin et al. 2020).

Table 5
Monophthongs incorrectly predicted to optionally round after labials

		PASSM	IDENTP			PASSM	IDENTP
<i>Pae</i>		10	10	<i>Pa</i>		10	10
		$\mathcal{H}$				$\mathcal{H}$	
 0.5	i. <i>Pae</i>	10		10	 0.5	i. <i>Pa</i>	10
 0.5	ii. <i>Poe</i>		10	10	 0.5	ii. <i>Po</i>	10

Table 6
Monophthongs incorrectly predicted to optionally round after labials

		PASSM	IDENTP			PASSM	IDENTP
<i>Pae</i>		1	10	<i>Pa</i>		1	10
		$\mathcal{H}$				$\mathcal{H}$	
 0.5	i. <i>Pae</i>	1		1	 0.5	i. <i>Pa</i>	1
 0.5	ii. <i>Poe</i>		10	10	 0.5	ii. <i>Po</i>	10

Building directly on Garvin, Lapierre, and Inkelas's (2018) treatment of geminates, $t\tau = (t^1 t^2 \cdot t^2 t^3)$, I propose that A'ingae diphthongs consist of four q's. The first two q's correspond to the first target of the diphthong; the other two q's correspond to the second target. Thus, for example, $ae = (a^1 a^2 e^2 e^3)$ and $oe = (o^1 o^2 e^2 e^3)$. This correctly predicts that while A'ingae diphthongs are longer than monophthongs, one component of a diphthong is shorter than a monophthong. Diphthongs arising in morphologically complex words involve two monophthongal vowels in the input. I assume that the two vowels fuse into one and undergo a reduction of the transitional ($q^3 q^1$) subsegments (e.g., $a + e = (a^1 a^2 a^3)(e^1 e^2 e^3) \rightarrow ae = (a^1 a^2 e^2 e^3)$), along with or before postlabial rounding. Similar reduction of transitional subsegments has been previously proposed by Inkelas and Shih (2017:14) for Polish singly articulated geminate affricates (e.g., $(t^1 t^2 \epsilon^3)(t^1 t^2 \epsilon^3) \rightarrow (t^1 t^2 t^2 \epsilon^3)$).

Furthermore, I assume that changing the feature of one subsegment (q) between the input and the output incurs only one-third, or $\frac{1}{3}$, of a violation of the respective IDENTITY constraint. Under this assumption, changing a feature of a monophthongal vowel (which consists of three q's) incurs a full violation ($3 \times \frac{1}{3} = 1$), but changing a feature of one vocalic target of a diphthong (which consists of two q's) incurs only two-thirds of a violation ($2 \times \frac{1}{3} = \frac{2}{3}$).² This predicts that a monophthongal vowel may surface faithfully, whereas the same vowel in a diphthong exhibits a TETU effect. More specifically, if PASSM has an appropriately low weight, the monophthongal /a/ will not round after labial consonants; rounding all three subsegments of a monophthong incurs sufficiently many IDENTITY violations to rule it out. This pattern matches the findings of Schwarz et al. (2019) and Garvin et al. (2020), who observe that the number of subsegments corresponds to vowel strength, resulting in the application or under-application of various phonological processes.

Additionally, I assume that the first subsegment (q^1 —the transition into a vowel) and the last subsegment (q^3 —the transition out of a vowel) must match the quality of the adjacent subsegment they transition into or out of. I formalize this with the *LONE-q constraint (7). The *LONE-q constraint rules out outputs where only one or two of a monophthong's subsegments become rounded.

(7) *LONE-q

Assign a violation for each subsegment that is not adjacent to an identical subsegment.

Finally, I recast LABIALASSIMILATION (5) as a subsegmental correspondence constraint couched in Agreement by Projection (Hansson 2014), a refinement of Agreement by Correspondence (Hansson 2001,

² Alternatively, the IDENTITY constraints could be defined as counting violations of subsegmental q's, as opposed to full segments. If this analysis were adopted, the weight of IDENTITY[LABIAL] would be correspondingly lower, but the core mechanics of the account would remain unchanged.

Rose and Walker 2004, Shih and Inkelas 2019), which models “spreading” as featural agreement. Specifically, I use $*[+LABIAL][-LABIAL]/c::v$ (8), which requires that a vocalic subsegment (v) have the same value for labiality as its preceding consonantal subsegment (c). For a similar account of vowel breaking in Huave (Kim 2008), see Inkelas and Shih 2017.

$$(8) \quad * [+LABIAL] [-LABIAL] / c :: v \quad (*+P-P_{c::v})$$

Assign a violation mark for each consonantal subsegment (c) immediately followed by a vocalic subsegment (v) with a mismatched $[\pm Labial]$ value.

The tableau capturing the fact that monophthongs do not round after labial vowels is given in table 7. The capital “P” in the input and the output candidates stands for an internally uniform labial consonant—that is, $P = (p^1 \ p^2 \ p^3)$. The (approximated) constraint weights were found using the Maxent Grammar Tool (Hayes 2008).

The diphthong /ae/, on the other hand, can optionally undergo rounding when preceded by a labial consonant, because its first portion has only two subsegments. This represents a case of TETU (McCarthy and Prince 1994)—with fewer subsegments come fewer faithfulness (IDENTITY) violations, making it possible for the unmarked structure $(*+P-P_{c::v})$ to emerge, where the labial features of the consonant and the first part of the diphthong match (table 8). (Since the actual counts of rounded and unrounded realizations of /ae/ are unavailable, the constraint weights were found assuming that they are equally likely.)

Table 7

Monophthongs correctly predicted not to round after labials

	P ($a^1 \ a^2 \ a^3$)	$*+P-P_{c::v}$ 12.5	IDENTP 18.8	$*LONE-q$ 14.4	$\mathcal{H}$
$\mathbb{E}^{\infty}$ i.	P ($a^1 \ a^2 \ a^3$)	1			12.5
ii.	P ($o^1 \ a^2 \ a^3$)		$\frac{1}{3}$	1	20.6
iii.	P ($o^1 \ o^2 \ a^3$)		$\frac{2}{3}$	1	26.9
iv.	P ($o^1 \ o^2 \ o^3$)		1		18.8

Table 8

Diphthongs correctly predicted to optionally round after labials

	P ($a^1 \ a^2 \ e^2 \ e^3$)	$*+P-P_{c::v}$ 12.5	IDENTP 18.8	$*LONE-q$ 14.4	$\mathcal{H}$
$\mathbb{E}^{\infty}_{0.5}$ i.	P ($a^1 \ a^2 \ e^2 \ e^3$)	1			12.5
ii.	P ($o^1 \ a^2 \ e^2 \ e^3$)		$\frac{1}{3}$	2	35.0
$\mathbb{E}^{\infty}_{0.5}$ iii.	P ($o^1 \ o^2 \ e^2 \ e^3$)		$\frac{2}{3}$		12.5
iv.	P ($o^1 \ o^2 \ o^2 \ e^3$)		1	1	33.2
v.	P ($o^1 \ o^2 \ o^2 \ o^3$)		$1\frac{1}{3}$		25.0

Table 9

Diphthongs correctly predicted not to round after nonlabials

	C ($a^1 a^2 e^2 e^3$)	*+P–P _{c::v} 12.5	IDENTP 18.8	*LONE-q 14.4	$\mathcal{H}$
i. C ($a^1 a^2 e^2 e^3$)					0.0
ii. C ($o^1 a^2 e^2 e^3$)		1	$\frac{1}{3}$	2	47.5
iii. C ($o^1 o^2 e^2 e^3$)		1	$\frac{2}{3}$		25.0
iv. C ($o^1 o^2 o^2 e^3$)		1	1	1	45.7
v. C ($o^1 o^2 o^2 o^3$)		1	$1\frac{1}{3}$		37.5

Finally, *+P–P_{c::v} penalizes labialization when /ae/ is preceded by a nonlabial onset. As a result, the diphthong never rounds after coronals, velars, and the like (table 9).

To summarize the discussion so far, adopting Q-Theory’s sub-segmental representations captures the fact that while the diphthongal /ae/ undergoes optional rounding after labial consonants, the monophthongal /a/ always surfaces faithfully.

5 Diachrony of Other Diphthongs

The Q-Theoretic analysis accounts for the fact that a phonological process aimed at markedness reduction (such as postlabial rounding) may target diphthongs, while monophthongs remain unaffected. In contemporary A’ingae, postlabial rounding applies only to /ae/. However, the language’s diachrony reveals that other diphthongs also used to be affected.

Drawing on data from morphologically complex words and borrowings, in other work (Dąbkowski 2023) I have identified a diachronic *postlabial raising*, whereby postlabial *ai developed into contemporary ii. I propose that the development of postlabial raising took place in two steps (9). First, postlabial *ai rounded to *oi (i.e., *ai > *oi / P₋) (9i). Second, postlabial *oi unrounded to ii (i.e., *oi > ii / P₋) (9ii).³

(9) Postlabial raising

0. *koehefa-ite	*ta?va-ite	waita (Kichwa)	baila (Spanish)
i. *koehefoite	*tavoite	*(rosa)voita	^m boira
ii. koehefiite	taviite	(rosa)viita	^m biira
summer-PRD	cotton-PRD	calendula	dance

(A’ingae and Spanish data from Borman 1976, Dąbkowski 2023; Kichwa data from Chango A. and Potosí C. 2009, ALDP 2018)

³ In Dąbkowski 2023, I propose a different trajectory, where the postlabial *ai first rounded to *ui (*ai > *ui / P₋) and then *u unconditionally unrounded to i (*u > i). Either trajectory involves a postlabial rounding of *ai to the exclusion of *a, supporting the main claim of the present work.

Both sound changes are directly attested. First, the postlabial rounding of *ai* is observed in one borrowing from Spanish, reported in Borman's (1976) dictionary with *oi*. Additionally, there is one instance of postlabial rounding of *ei* (10i). (For comparison, non-postlabial diphthongs (e.g., *ai* in *airo* 'mountain') have not been affected.)

(10) *Step I: Postlabial rounding of ai, ei*

0. Source	<i>baila</i> (Spanish)	<i>peineta</i> (Spanish)	<i>airo</i> (Secoya)
i. Borman	^m <i>boira</i>	^m <i>bōĩnēta</i>	<i>airo</i>
	dance	comb	mountain

(A'ingae data from Borman 1976; Secoya data from ALDP 2018)

Second, all the words reported in Borman 1976 with a postlabial *oi* are nowadays realized with *ii* (11ii).⁴ (Non-postlabial *oi*, as in *sōĩ* 'tamal', has not been affected.) The phonetic motivation for postlabial unrounding is not clear. I suggest it may be construed as centralization, reflecting a crosslinguistic affinity between labial consonants and central vowels (Vallée, Rossato, and Rousset 2009), or as an instance of hypercorrective dissimilation.

(11) *Step II: Postlabial unrounding of oi*

i. Borman	<i>pōĩnē</i>	<i>p^hoikā</i>	^m <i>boira</i>	<i>mōĩte</i>	<i>sōĩ</i>
ii. Present	<i>pīĩnē</i>	<i>p^hiikā</i>	^m <i>biira</i>	<i>mīĩte</i>	<i>sōĩ</i>
	both	glimpse	dance	when.not	tamal

Postlabial rounding (**ai* > **oi*) and unrounding (**oi* > *ii*) both applied only to diphthongs; postlabial monophthongs (e.g., in *pa* 'die', *p^horo* 'be touching', ^m*baʔve* 'more or less', ^m*bo* 'gather', *fōpnō* 'skirt', *va* 'this', and *mēĩfo* 'not have') were unaffected.

The postlabial rounding (or the telescoped postlabial raising) of **ai* is not a productive phonological process. First, postlabial raising sustains lexical exceptions, such as *paiĩsi* 'paiche'. Second, postlabial raising does not apply regularly to new derivatives (although it has been acquired as an optional derived-environment rule by some). Third, many historically complex forms with *ii*, such as *koehefiĩte* 'summer-PRD' and *taviĩte* 'cotton-PRD' (9ii), have been, for some speakers, leveled back to *koehefaite*, *tavaite*. (For more information on postlabial raising, see Dąbkowski 2023.)

Nevertheless, the rounding of **ai* as well the subsequent unrounding of **oi* affected postlabial diphthongs, to the exclusion of the monophthongal **a* and **o*. Since sound changes are the result of once-active phonetic and phonological pressures, they provide additional evidence that A'ingae has—also in its past—shown less faithfulness to one component of a diphthong than to the corresponding monophthong, further strengthening the case for the representational apparatus of Q-Theory, which predicts this exact asymmetry.

⁴ Borman worked in the Cofán communities beginning in 1954 (Hugo Lucitante, pers. comm.). Therefore, the postlabial unrounding of *oi* must have taken place in the past 50–70 years.

6 Conclusion

In this squib, I have described and analyzed the process of postlabial rounding in A'ingae. After labials, the diphthongal /ae/ may optionally surface as [oe], but the monophthongal /a/ always surfaces as [a]. Similar asymmetries can be observed in the language's history: postlabial *ai rounded to *oi, and postlabial *oi centralized to ii. Both sound changes affected only diphthongs, to the exclusion of postlabial *a and *o (although they did not give rise to active phonological processes). Thus, we observe a recurring pattern where the language's diphthongs—but not monophthongs—are prone to assimilation and centralization/dissimilation.

To account for the systematic difference between monophthongs and diphthongs, I have adopted the subsegmental representations of Q-Theory. A monophthong is longer and consists of more subsegments than one component of a diphthong. Thus, unfaithfulness to the features of the latter incurs fewer IDENTITY violations, creating conditions in which unmarked structures (such as labial assimilation) may be observed to emerge. In this way, the A'ingae patterns bear out a prediction of Q-Theory.

Finally, by allowing subsegments to incur partial constraint violations, I have demonstrated that Q-Theory has some of the same capacity for modeling gradient phonology as the Gradient Symbolic Representations framework (e.g., Rosen 2016, Smolensky and Goldrick 2016). The subsegments of Q-Theory, however, are phonetically motivated and discrete, making it the more restrictive of the two frameworks.

References

- ALDP. 2018. *Mostly Spanish borrowings mostly from Borman dictionary*. Brown University, A'ingae Language Documentation Project's unpublished database.
- AnderBois, Scott, Nicholas Emlen, Hugo Lucitante, Chelsea Sanker, and Wilson de Lima Silva. 2019. Investigating linguistic links between the Amazon and the Andes: The case of A'ingae. Paper presented at the 9th Conference on Indigenous Languages of Latin America, University of Texas at Austin.
- Borman, Marlytte "Bubs". 1962. Cofán phonemes. In *Studies in Ecuadorian Indian languages*, ed. by Benjamin Franklin Elson, 45–59. Instituto Lingüístico de Verano (Summer Institute of Linguistics). <https://www.sil.org/resources/archives/8877>.
- Borman, Marlytte "Bubs". 1976. *Vocabulario cofán: Cofán–castellano, castellano–cofán*. Instituto Lingüístico de Verano (Summer Institute of Linguistics). <https://www.sil.org/resources/archives/10958>.
- Brandt, Adam. 2024. Stress and nasality as conditioning factors for 'free' variation in A'ingae vowels. Talk presented at the 17th Toronto Undergraduate Linguistics Conference, University of Toronto.

- Chango A., María T., and Fabián Potosí C., eds. 2009. *Kichwa: Yachakukkunapa Shimiyuk Kamu*. Ministerio de Educación, Ecuador.
- Dąbkowski, Maksymilian. 2020. A'ingae field materials. California Language Archive, Survey of California and Other Indian Languages, University of California, Berkeley. <https://doi.org/10.7297/X2HH6HKG>.
- Dąbkowski, Maksymilian. 2021a. A'ingae (Ecuador and Colombia) – Language snapshot. *Language Documentation and Description* 20:1–12. <https://doi.org/10.25894/ldd28>.
- Dąbkowski, Maksymilian. 2021b. Dominance is non-representational: Evidence from A'ingae verbal stress. *Phonology* 38:611–650. <https://doi.org/10.1017/S0952675721000348>.
- Dąbkowski, Maksymilian. 2023. Postlabial raising and paradigmatic leveling in A'ingae: A diachronic study from the field. In *Proceedings of the Linguistic Society of America* 8(1), ed. by Patrick Farrell. Linguistic Society of America. <https://doi.org/10.3765/plsa.v8i1.5428>.
- Dąbkowski, Maksymilian. 2024a. The phonology of A'ingae. *Language and Linguistics Compass* 18:e12512. <https://compass.onlinelibrary.wiley.com/doi/abs/10.1111/lnc3.12512>.
- Dąbkowski, Maksymilian. 2024b. Two grammars of A'ingae glottalization: A case for Cophonologies by Phase. *Natural Language and Linguistic Theory* 42:437–491. <https://doi.org/10.1007/s11049-023-09574-5>.
- Dąbkowski, Maksymilian. To appear. Phasal strength in A'ingae classifying subordination. In *Proceedings of the 2023/2024 Annual Meeting on Phonology*. Linguistic Society of America. <https://ling.auf.net/lingbuzz/007821>.
- Fischer, Rafael, and Kees Hengeveld. 2023. A'ingae (Cofán/Kofán). In *Language isolates 1: Aikanā to Kandozi-Shapra*, ed. by Patience Epps and Lev Michael, 65–124. De Gruyter Mouton.
- Galloway, Brent Douglas. 1990. *Phonology, morphology, and classified word list for the Samish dialect of Straits Salish*. University of Ottawa Press.
- Garvin, Karee, Myriam Lapierre, and Sharon Inkelas. 2018. A Q-Theoretic approach to distinctive subsegmental timing. In *Proceedings of the Linguistic Society of America* 3, ed. by Patrick Farrell. Linguistic Society of America. <https://journals.linguisticsociety.org/proceedings/index.php/PLSA/article/view/4330>.
- Garvin, Karee, Myriam Lapierre, Martha Schwarz, and Sharon Inkelas. 2020. Modeling vowel quantity scales in Q Theory. In *Proceedings of the 2019 Annual Meeting on Phonology*, ed. by Hyunah Baek, Chikako Takahashi, and Alex Hong-Lun Yeung. Linguistic Society of America. <https://doi.org/10.3765/amp.v8i0.4685>.
- Hammarström, Harald, Robert Forkel, Martin Haspelmath, and Sebastian Bank, eds. 2020. *Glottolog 4.3 – Cofán*. Max Planck Insti-

- tute for the Science of Human History. <https://glottolog.org/resource/languoid/id/cofa1242#refs>.
- Hansson, Gunnar Ólafur. 2001. Theoretical and typological issues in consonant harmony. Doctoral dissertation, University of California, Berkeley. <https://www.proquest.com/dissertations-theses/theoretical-typological-issues-consonant-harmony/docview/304683446/se-2>.
- Hansson, Gunnar Ólafur. 2014. (Dis)Agreement by (Non)Correspondence: Inspecting the foundations. UC Berkeley Phonology Lab Annual Report: ABC Conference Archive. <http://linguistics.berkeley.edu/phonlab/documents/2014/ABCC/Hansson.pdf>.
- Hayes, Bruce. 2008. Maxent Grammar Tool. <https://linguistics.ucla.edu/people/hayes/MaxentGrammarTool/>.
- Inkelas, Sharon, and Stephanie S. Shih. 2016. Re-representing phonology: Consequences of Q Theory. In *NELS 46*, ed. by Christopher Hammerly and Brandon Prickett, 161–174. University of Massachusetts Amherst, Graduate Linguistics Students Association.
- Inkelas, Sharon, and Stephanie S. Shih. 2017. Looking into segments. In *Proceedings of the 2016 Annual Meeting on Phonology*, ed. by Karen Jesney, Charlie O’Hara, Caitlin Smith, and Rachel Walker. Linguistic Society of America. <https://doi.org/10.3765/amp.v4i0.3996>.
- Kim, Yuni. 2008. Topics in the phonology and morphology of San Francisco del Mar Huave. Doctoral dissertation, University of California, Berkeley. <https://www.proquest.com/dissertations-theses/topics-phonology-morphology-san-francisco-del-mar/docview/304693674/se-2>.
- Lakshmi, V. Swarajya. 1982. *A descriptive grammar of Cuddapah dialect*. Telugu Akademi.
- Lionnet, Florian. 2017. A theory of subfeatural representations: The case of rounding harmony in Laal. *Phonology* 34:523–564.
- McCarthy, John J., and Alan Prince. 1994. The emergence of the unmarked: Optimality in prosodic morphology. In *NELS 24*, ed. by Mercè González, 2:333–379. University of Massachusetts Amherst, Graduate Linguistics Students Association.
- Repetti-Ludlow, Chiara. 2021. The A’ingae co-occurrence constraint. In *Proceedings of the 2020 Annual Meeting on Phonology: Supplemental proceedings*, ed. by Ryan Bennett, Richard Bibbs, Mykel Loren Brinkerhoff, Max J. Kaplan, Stephanie Rich, Amanda Rysling, Nicholas Van Handel, and Maya Wax Cavallaro. Linguistic Society of America. <https://doi.org/10.3765/amp.v9i0.4859>.
- Repetti-Ludlow, Chiara, Haoru Zhang, Hugo Lucitante, Scott Ander-Bois, and Chelsea Sanker. 2019. A’ingae (Cofán). *Journal of the International Phonetic Association: Illustrations of the IPA* 1–14.
- Rivet, Paul. 1924. *Langues de l’Amérique du Sud et des Antilles*. Société Linguistique de Paris.

- Rivet, Paul. 1952. Affinités du Kofán. *Anthropos* 47:203–234.
- Rose, Sharon, and Rachel Walker. 2004. A typology of consonant agreement as correspondence. *Language* 80:475–531. <https://www.jstor.org/stable/4489721>.
- Rosen, Eric. 2016. Predicting the unpredictable: Capturing the apparent semi-regularity of rendaku voicing in Japanese through Gradient Symbolic Computation. In *Proceedings of the 42nd Annual Meeting of the Berkeley Linguistics Society*, ed. by Emily Clem, Geoff Bacon, Andrew Cheng, Virginia Dawson, Erik Hans Maier, Alice Shen, and Amalia Horan Skilton, 235–249. University of California, Berkeley, Berkeley Linguistics Society.
- Ruhlen, Merritt. 1987. *A guide to the world's languages*. Vol. 1, *Classification*. Stanford University Press. <https://books.google.com/books?id=mYwmDE3f6wUC>.
- Sanker, Chelsea, and Scott AnderBois. 2024. Reconstruction of nasality and other aspects of A'ingae phonology. *Cadernos de Etnolingüística* 11:e110101. <http://www.etnolingustica.org/article:vol11n1>.
- Schwarz, Martha, Myriam Lapierre, Karee Garvin, and Sharon Inkelas. 2019. Recent advances in Q Theory: Segment strength. In *Proceedings of the Linguistic Society of America 4*, ed. by Patrick Farrell. Linguistic Society of America. <https://journals.linguistic-society.org/proceedings/index.php/PLSA/article/view/4550>.
- Shih, Stephanie S., and Sharon Inkelas. 2019. Autosegmental aims in surface-optimizing phonology. *Linguistic Inquiry* 50:137–196. https://doi.org/10.1162/ling_a_00304.
- Smolensky, Paul, and Matthew Goldrick. 2016. Gradient Symbolic Representations in grammar: The case of French liaison. Rutgers Optimality Archive 1286. <http://roa.rutgers.edu/>.
- Smolensky, Paul, and Géraldine Legendre. 2006. *The harmonic mind*. Vol. 1, *From neural computation to Optimality-Theoretic grammar*. MIT Press. <https://books.google.pl/books?id=KlsiMwEACAAJ>.
- Vallée, Nathalie, Solange Rossato, and Isabelle Rousset. 2009. Favoured syllabic patterns in the world's languages and sensorimotor constraints. In *Approaches to phonological complexity*, ed. by François Pellegrino, Egidio Marsico, Ioana Chitoran, and Christophe Coupé, 11–39. De Gruyter Mouton. <https://doi.org/10.1515/9783110223958>.

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